

Mahatma Education Society's

Pillai HOC College of Engineering and Technology, Rasayani (Autonomous)

Department of Computer Application (MCA)

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Literature Survey

Group No.: RPG10

Title of Research Paper: BlockVote-A Decentralized Voting System ensuring Integrity, Transparency & Trust

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Paper 1: E Voting via Upgradable Smart Contracts on Blockchain

Authors: Mohammad Saim, Ilma Shah, Mohd Mamoon, Abdus Samad Published: 2022 International Conference on Futuristic Technologies (INCOFT) | ISBN: 978 1 6654 5046 1

EXPLANATION

This paper proposes an electronic voting system built on the Ethereum blockchain to enhance privacy, authenticity, and verifiability in the electoral process. To overcome the traditional rigidity of immutable blockchain deployments, the central innovation of this system is the integration of upgradable smart contracts using the OpenZeppelin proxy pattern plugin. This architecture separates the smart contract into a Proxy contract that holds the state variables and an Implementation contract that contains the execution logic, allowing election managers to update application logic and fix bugs via delegate calls without losing state data or changing the main contract address. On the client side, the system features a ReactJS web application connected via Web3.js, requiring voters to verify their identity through Aadhaar details and facial recognition via a Python Flask backend before casting a ballot through MetaMask. Furthermore, transaction costs are managed using a Gas Station Network setup, which acts as a decentralized group of relayers to handle gas fees and prevent financial burdens on individual voters during vote casting.

ADVANTAGES

- Deploying upgradable proxy smart contracts allows election organizers to update application logic and patch bugs iteratively without altering the contract address or losing state variables.
- Integrating a Gas Station Network abstracts transaction costs away from voters, easing user onboarding and removing the financial burden of purchasing cryptocurrency.

LIMITATIONS

- Proxy configuration introduces centralization risks if the admin key is compromised.
- Centralized Flask server becomes a processing bottleneck for facial recognition.

IMPROVEMENTS

- Uses OpenZeppelin UUPS Proxy Pattern so contract logic can be upgraded without losing blockchain state, and no single admin key controls upgrades.
- Implements a custom Relay Wallet (gas station model) on the server, allowing voters to submit votes through a standard web form with zero blockchain interaction.

Paper 2: Secured Electronic voting system using Blockchain Technology

Authors: S. Gnanapriya, M Ram Eshwar, CG Shrikhiran, J Vijay Chandar, V. Nandhakumar

Published: 2023 International Conference on Research Methodologies in Knowledge Management, Artificial Intelligence and Telecommunication Engineering (RMKMATE) | ISBN: 979 8 3503 0570 8

EXPLANATION

This paper explores the application of Ethereum smart contracts to build a transparent and tamper proof electronic voting system that addresses security flaws in traditional voting methods. The proposed architecture uses self executing Solidity smart contracts to automate the voting process and record each vote as an immutable transaction on the Ethereum blockchain. To optimize user experience and client side performance, the application frontend is developed using the React JS framework, which provides superior rendering speed and component reusability compared to alternative frameworks like Angular JS. Voters interact with the system by connecting their Web3 digital wallet via MetaMask, where valid account addresses are searched against registered voters to grant ballot access. Once a voter approves the transaction in their wallet, the vote is broadcast across the peer to peer network and permanently added to the ledger upon reaching consensus.

ADVANTAGES

- Developing the client interface with React JS significantly improves frontend loading efficiency and system responsiveness through dynamic reusable components.
- The use of Ethereum smart contracts ensures automated and accurate vote tallying while encrypting ballot data to prevent retroactive manipulation.

LIMITATIONS

- The architecture depends on a centralized admin whitelist for voter validation, which introduces a central point of failure during registration.
- The system requires users to hold Ether for transaction fees via MetaMask, creating a financial and technical barrier for everyday voters.

IMPROVEMENTS

- Implements organization scoped voter management, where each organization independently manages its own voter roster through email based registration (MongoDB), removing dependency on a single global admin whitelist.
- Built on Next.js 15 with TanStack Query, where all blockchain data is served through REST APIs, allowing voters to use a standard browser interface without requiring a MetaMask extension or an ETH wallet.

Paper 3: A Secure and Transparent Blockchain driven E-Voting System using Smart Contract and IPFS-based Scalability

Authors: Saptaparnee Saha, Santanu Shaw, Subhas Barman, Jhuma Dutta, Argha Sinha

Published: 2025 IEEE Guwahati Subsection Conference (GCON) | ISBN: 979-8-3315-1345-0

EXPLANATION

This paper presents an Ethereum-based electronic voting system that combines Solidity smart contracts with off-chain storage to solve the high gas costs and scalability constraints of on-chain elections. During voter registration, user identity is verified through a multi-factor authentication mechanism involving the UIDAI Aadhaar API and Twilio-powered OTP verification. To ensure data privacy and reduce network congestion, voter metadata is serialized into JSON, encrypted using AES-256 in CBC mode, hashed with SHA-256 for integrity, and saved on the InterPlanetary File System (IPFS) using a Merkle DAG structure. The smart contract records only the resulting Content Identifier (CID) alongside a pseudonymous Digital Voter ID on the blockchain ledger. The architecture also features a post-election statistics module that extracts demographic trends from IPFS and event logs, while static code analysis via the Oyente tool confirms that the smart contracts are safe from vulnerabilities such as reentrancy and integer overflow.

ADVANTAGES

- Quantitative performance evaluations show high system efficiency, achieving swift vote-casting latency alongside substantial processing throughput.
- Offloading encrypted voter metadata to the InterPlanetary File System drastically reduces gas consumption, requiring significantly fewer gas units for the entire voting process compared to traditional full on-chain implementations.

LIMITATIONS

- Data availability relies heavily on consistent cluster maintenance and active node pinning, where a failure in node persistence could lead to broken content identifier links or unretrievable off-chain records.
- The framework does not currently guarantee complete uncoercibility, leaving voters potentially vulnerable to external pressure unless advanced vote-mixing methods such as zero-knowledge proofs or blind signatures are integrated.

IMPROVEMENTS

- Uses Pinata for persistent, authenticated IPFS pinning, ensuring biometric metadata CIDs remain accessible without relying on ephemeral public nodes.
- Protects voter identity using keccak256 nullifier hashes (orgSlug + email + server secret), ensuring no personally identifiable information is ever stored on chain or in IPFS.

Paper 4: Blockchain based Electronic Voting System Design with Smart Contracts

Authors: Wan Auzan Bin W Abdulah, Syed Farid Syed Adnan Published: 2023 IEEE Symposium on Computers & Informatics (ISCI) | ISBN: 979 8 3503 0236 3

EXPLANATION

The paper introduces a decentralized electronic voting framework developed on the Ethereum blockchain using Solidity smart contracts to ensure fairness, transparency, and immutability during elections. Built using the Truffle development environment and tested on a local Ganache network, the system relies on MetaMask for user account verification and transaction processing. Before casting a ballot, voters and candidates complete an initial registration step where legitimate users receive unique cryptographic addresses and pre allocated test ethers enabling exactly one transaction to prevent double voting. A key focus of the study is a detailed transaction cost analysis examining how smart contract code complexity, candidate list sizes, and data type choices directly impact gas fees on the Ethereum Virtual Machine.

ADVANTAGES

- Outlines practical smart contract optimization techniques that demonstrate how simplifying code logic and reducing storage loops effectively lowers gas consumption.
- Enforces strict vote integrity by binding each voter account to a single transaction limit verified through the network.

LIMITATIONS

- The system remains highly sensitive to Ethereum gas fee fluctuations, which makes deployment and voting costs unpredictable during periods of heavy network congestion.
- Client side security is largely unaddressed, leaving voters remote devices vulnerable to malware, keyloggers, or phishing attacks during transaction signing.

IMPROVEMENTS

- Minimizes gas consumption by using bytes32 election IDs and O(1) mapping lookups throughout the contract, avoiding dynamic arrays and expensive storage patterns.
- Uses Hardhat deployment scripts (deployProxy.js, deployV2Impl.js) for automated, reproducible contract deployment, eliminating manual steps that introduce human error or security gaps.

Paper 5: Unleashing Innovation: Experimental Evaluation of Blockchain Assisted Electronic Voting System using Secured Authentication Scheme

Authors: K. Kalaiselvi, T. Venkatesan, K. Saravanan, S. Ravi, Shalini M, J.Karthi Published: 2024
International Conference on Intelligent Systems for Cybersecurity (ISCS) | ISBN: 979 8 3503 7523 7

EXPLANATION

This research paper explores the development and experimental evaluation of a Blockchain Assisted Electronic Voting System designed to increase security, transparency, and public trust in electronic voting. The system architecture encompasses a user facing client interface, a decentralized blockchain network utilizing platforms like Ethereum or Hyperledger Fabric, an authentication module, and a smart contract driven voting mechanism. To verify voter identity while safeguarding data confidentiality, the framework employs a multi factor authentication scheme combining biometric parameters such as facial or fingerprint recognition, public private cryptographic key pairs, and digital signatures. The authors conduct empirical testing evaluating false acceptance rates, false rejection rates, and overall accuracy, demonstrating that their proposed authentication scheme outperforms baseline methods by achieving a high overall verification accuracy alongside minimal false acceptance risks.

ADVANTAGES

- Achieves a high overall authentication accuracy of 99.3 percent while maintaining a minimal false acceptance rate of 0.1 percent across test instances.
- Employs distributed ledger immutability and cryptographic digital signatures to ensure that once a vote is cast, it cannot be altered or deleted by external parties.
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LIMITATIONS

- The strict multi factor authentication enforcement leads to a higher false rejection rate, which may inadvertently deny access to legitimate voters during authentication.
- The complex cryptographic workflow involved in managing key pairs and multi factor steps may create usability hurdles that reduce overall user trust and adoption.

IMPROVEMENTS

- Replaces complex cryptographic key-pair MFA with Email OTP, providing a familiar, zero-configuration authentication step that voters can complete without technical knowledge.
- Contract upgrades are governed by a 2-of-3 Guardian multi-signature mechanism, preventing any single authority from pushing changes and reducing governance abuse without adding cryptographic complexity for voters.

Paper 6: BP Vot: Blockchain Based e Voting Using Smart Contracts, Differential Privacy, and Self Sovereign Identities

Authors: Hamza Baniata, Giovanni Caluna Published: 2025 IEEE Access | ISSN: 2169 3536

EXPLANATION

This paper introduces BP Vot, a privacy preserving electronic voting framework that integrates Ethereum smart contracts, Self Sovereign Identities, and a novel differential privacy mechanism. Deployed on a permissioned Hyperledger Besu network, the system utilizes Self Sovereign Identity standards to allow voters to cast ballots using Web3 digital identity wallets and verifiable pseudonymous credentials without exposing real personal information. To hide voter candidate preferences on the public ledger, the smart contract implements a differential privacy mechanism that randomly pivots votes from a selected candidate to others during the voting window. Once the election closes, the final tally is statistically approximated to determine the election outcomes while maintaining mathematical robustness against reconstruction attacks and complying with privacy laws like GDPR.

ADVANTAGES

- Provides a 24 percent enhancement in transaction latency compared to existing state of the art solutions by processing transactions in approximately one second.
- Ensures strong privacy guarantees and compliance with regulations by mathematically obscuring direct voter to candidate mappings on the distributed ledger.

LIMITATIONS

- The system produces statistical approximations of election outcomes rather than exact, deterministic vote counts.
- Approximation accuracy decreases as the total candidate count increases due to the inverse relationship between candidate numbers and the differential privacy threshold.

IMPROVEMENTS

- Records exact, tamper proof vote counts on chain through the castVoteRelayed function, ensuring precise results regardless of candidate count.
- A pre flight cache layer (MongoDB) validates each vote attempt before it reaches the blockchain, instantly rejecting duplicate or ineligible submissions without wasting gas.

Paper 7: Securing Blockchain based E Voting through Shamir's Secret Sharing over Ethereum

Authors: Esma Beydili, Umut Can Cabuk, Gokhan Dalkilic, Yusuf Ozturk Published: 2025

IEEE | ISBN: 979 8 3315 0993 4

EXPLANATION This paper addresses the privacy vulnerabilities caused by the inherent transparency of public smart contracts by integrating a cryptographic secret sharing algorithm into an Ethereum based electronic voting system. To ensure that voter choices remain completely confidential until the election concludes, the framework fragments the voting data into multiple secret shares that are distributed among participants. The system requires a predefined quorum or threshold of these fragments to be combined using Lagrange interpolation before the original secret can be reconstructed and the final election results revealed. Developed and tested on the Sepolia network, the prototype requires voters to submit their assigned secret shares alongside their ballot, which successfully obscures candidate selection and prevents double voting while ensuring that no single central entity can prematurely access the election data.

ADVANTAGES

- Successfully obfuscates candidate selection and drastically improves ballot anonymity by fragmenting the vote data using cryptographic secret sharing.
- Prevents major electoral errors and premature result leaks by enforcing multi party authorization and quorum thresholds for secret reconstruction.

LIMITATIONS

- The integration of the secret sharing algorithm introduces high computational complexity during the final result generation phase.
- The increased processing time and algorithmic overhead directly affect system accessibility by causing noticeable delays for voters attempting to access the platform and cast their ballots.

IMPROVEMENTS

- Uses an on chain hasVoted mapping for constant time double vote detection, with results computed directly from vote count fields, eliminating reconstruction overhead regardless of voter count.
- A NonceManager with a serialized relay queue safely handles concurrent vote submissions without transaction collisions, maintaining consistent response time under heavy user load.

Paper 8: Digital Voting with Blockchain using Interplanetary File System and Practical Byzantine Fault Tolerance

Authors: Giddaluru Somasekhar, Sreedhar Jinka, Chinna Kullayappa Kanekal, Anusha Marouthu Published: Engineering, Technology & Applied Science Research, Vol. 14, No. 6, 2024

EXPLANATION

This paper presents a digital voting system that combines blockchain technology with the InterPlanetary File System (IPFS) and the practical Byzantine Fault Tolerance (pBFT) consensus algorithm. The architecture decentralizes the voting process by securely storing voting data off-chain in IPFS through distinctive hashes, which are then linked to corresponding immutable blockchain records to ensure data authenticity. To optimize performance and reduce transaction expenses, the system employs advanced smart contract coding techniques such as batching, caching, and efficient data packing to minimize gas costs. Furthermore, the pBFT algorithm ensures swift and reliable validation of each vote, overcoming node failures and mitigating malicious activities within the distributed network.

ADVANTAGES

- The integration of basic gas optimization techniques, such as batching and caching within the smart contracts, reduces average gas costs by up to 33 percent compared to existing enterprise blockchain models.
- The use of the practical Byzantine Fault Tolerance consensus algorithm allows the network to process a high volume of transactions rapidly, decreasing average latency by up to 55 percent and significantly reducing authentication delays.

LIMITATIONS

- High infrastructure maintenance overhead for running a permissioned PBFT network.
- Requires continuous legal and technical updates.

IMPROVEMENTS

- Uses Ethereum's public blockchain (Sepolia Testnet/Mainnet) instead of a permissioned PBFT cluster, eliminating the need to host and maintain custom consensus infrastructure.
- Implements per-election voter registration (`isRegisteredVoter[electionId][nullifierHash]`), making each election self-contained without requiring system-wide reconfiguration or legal re-certification.

Paper 9: Blockchain-Powered E-Voting: A Novel Approach to Secure Voter Authentication, Online Voting and Election Automation

Authors: B Sujatha, Yarrampati Ganesh, N Leelavathy, R Tamilkodi, Srirama Venkatesh, Bayye Sandhya, Toleti Sai Kowshik Published: Indian Journal of Science and Technology (2024) | ISSN: 0974-5645

EXPLANATION

This paper proposes a decentralized electronic voting system leveraging the Ethereum blockchain and smart contracts to automate election processes such as scheduling, voter registration, and result declaration. A key feature of the framework is its multi-tiered authentication process, utilizing AI and ML-driven biometric and facial recognition alongside OTP verification to securely verify voter identities. To maintain data immutability and protect privacy, sensitive voter details and biometric data are stored on the InterPlanetary File System (IPFS) rather than a centralized server. The frontend is developed using the Django web framework, providing an accessible interface for administrators, volunteers, and voters, while ensuring that each cast ballot is safely recorded as a tamper-proof transaction on the distributed ledger.

ADVANTAGES

- Integrating biometric and facial recognition increases authentication accuracy to approximately 98 percent, significantly reducing the risk of impersonation compared to traditional password or OTP-based methods.
- The use of smart contracts eliminates manual tallying processes, accelerating result generation by over 95 percent while ensuring complete transparency of the final vote count.

LIMITATIONS

- Limited validation for mobile scalability.
- Requires heavy backend cryptographic computation.

IMPROVEMENTS

- Offloads face detection to AWS Rekognition, a managed cloud API that performs liveness, brightness, and multi-face quality checks across desktop and mobile browsers.
- Delivers the system as a browser-based Next.js application, requiring no native app installation and providing immediate access on smartphones and desktops without platform-specific builds.

Paper 10: Enhancing Security and Transparency in Online Voting through Blockchain Decentralization

Authors: Inderpreet Singh, Amandeep Kaur, Parul Agarwal, Sheikh Mohammad Idrees Published: SN Computer Science (2024) | ISSN/ISBN: DOI: 10.1007/s42979-024-03286-2

EXPLANATION

This paper presents an automated and decentralized online voting system built on the Ethereum blockchain and Solidity smart contracts to replace vulnerable, centralized e-government services. The system addresses major security, transparency, and data integrity challenges by eliminating intermediaries and recording voter transactions on an immutable distributed ledger. To protect sensitive citizen data such as personal identities and medical details from data leaks, the architecture restricts administrative access strictly to necessary information fields like contact details, names, and ages while using cryptographic techniques to secure sensitive biometric data. Furthermore, the system employs automated security checks, such as retina scan bytcodes matched against government datasets and OTP verifications, to authenticate voters before allowing them to cast immutable votes through a web application.

ADVANTAGES

- Replaces vulnerable centralized servers with a decentralized blockchain network, effectively eliminating single points of failure and preventing rogue insider data manipulation.
- Uses automated smart contracts to handle the election workflow from registration to result tabulation, significantly reducing manual errors, costs, and processing delays.

LIMITATIONS

- Standard non-upgradable contracts risk permanently embedding unexpected runtime logic flaws if unanticipated edge-cases occur during a live election.
- Relying on client-side browser frameworks exposes the voting submission loop to external web interface tampering by malicious actors.

IMPROVEMENTS

- Incorporates automated conditional checks (require / assert) that automatically execute state rollback exceptions to block invalid transaction mutations.
- Limits electoral administrative views strictly to segmented citizen data fields, shielding sensitive medical records from footprint tracking.